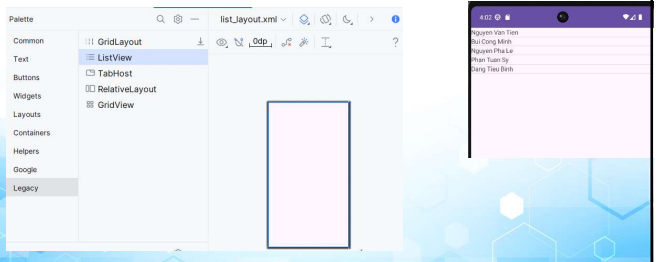
COS30017. SOFTWARE DEVELOPMENT FOR MOBILE DEVICES

Dr. Nguyen Dang Khoa
Email: kdnguyen@swin.edu.au
Phone: 0974.844.618

Week 7 – List



Legacy ➔ListView

Week 7 – List

Layout

- Create file list_layout
- Write LinearLayout
- Add ListView in LinearLayout ➔ Set ID to lsStudents

Kotlin code

- Create file List_Activity: AppCompatActivity()
- Override fun onCreate ➔ setContentView(R.layout.list_layout)
- Code to add infor to lsStudents

Layout

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".List_Activity">

    <ListView
        android:id="@+id/lsStudents"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Week 7 – List

```
public ArrayAdapter(
    android.content.Context context,
    @LayoutRes↗ int resource,
    T[] objects
)
```

Kotlin code

- **Defined lsStudents in Back-End**
- **Defined the string array for list of students**

```
lateinit var lsStudents: ListView
var strList: Array<String> = arrayOf("Nguyen Van Tien", "Bui Cong Minh",
"Nguyen Pha Le", "Phan Tuan Sy", "Dang Tieu Binh")
```

- **Binding data**

```
lsStudents= findViewById(R.id.lsStudents)
var arr = ArrayAdapter(this, R.layout.template_layout, strList)
lsStudents.setAdapter(arr)
```

➔ **Create a *template_layout layout***

Week 7 – List

```
public ArrayAdapter(
    android.content.Context context,
    @LayoutRes↗ int resource,
    T[] objects
)
```

template_layout.xml

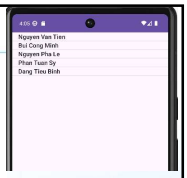
```
<?xml version="1.0" encoding="utf-8"?>

<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />
```

Week 7 – List

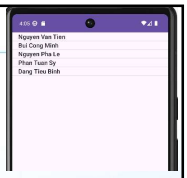
template_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
    <TextView xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ellipsize="marquee"
        android:singleLine="true"
        android:textAppearance="?android:attr/textAppearanceListItemSmall"
        android:paddingStart="16dp"
        android:paddingEnd="16dp"
        android:gravity="center_vertical" />
```



Week 7 – List - getItems

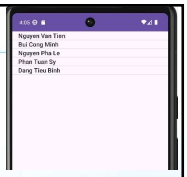
```
lsStudents.setOnItemClickListener({ parent, view, position, id ->
    val element = arr.getItem(position) as String
    Toast.makeText(this, element, Toast.LENGTH_SHORT).show()
})
```

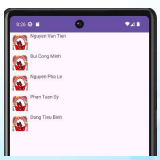


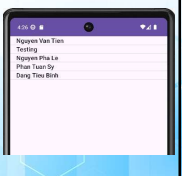
Week 7 – List - ChangeItems

➔ setAdapter AGAIN

```
strList.set(1,"Testing")
arr = ArrayAdapter(this, R.layout.template_layout, strList)
lsStudents.setAdapter(arr)
```







Week 7 – RecyclerView

```
@androidx.recyclerview.widget.RecyclerView
public void setAdapter(
    @Nullable RecyclerView.Adapter adapter
)
```

Layout

- Create file recyclerview_layout
- Add RecyclerView in LinearLayout ➔ Set ID to lsStudents

Kotlin code

- Create file Recycler_Activity: AppCompatActivity()
- Override fun onCreate ➔ setContentView(R.layout.recyclerview_layout)
- Code to add infor to lsStudents

Layout

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Recycler_Activity" >

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/lsStudents"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Similar with ListView

Week 7 – RecyclerView

```
@androidx.recyclerview.widget.RecyclerView
public void setAdapter(
    @Nullable RecyclerView.Adapter adapter
)
```

Kotlin code

- Defined lsStudents in Back-End
- Defined the string array for list of students

```
lateinit var lsStudents: RecyclerView
var strList: Array<String> = arrayOf("Nguyen Van Tien", "Bui Cong Minh",
"Nguyen Pha Le", "Phan Tuan Sy", "Dang Tieu Binh")
```

- Binding data

```
lsStudents= findViewById(R.id.lsStudents)
var arr = ArrayAdapter(this, R.layout.template_re_layout, strList)
lsStudents.setAdapter(arr)          //ERROR
```

➔ Create a *template_re_layout layout*

template_re_layout layout

```
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/txtName"    />
```

Similar with ListView

But need to set ID

Processing ERROR

- **Create class DataClass**

```
data class DataClass(var StudentNames:String)
```

- **Create class RecyclerViewAdapter**

Processing ERROR

- **Create class RecyclerViewAdapter**

```
class RecyclerViewAdapter (private val mList: List<DataClass>) : RecyclerView.Adapter<RecyclerViewAdapter.ViewHolderClass>() {

    override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): ViewHolderClass {
        var itemView = LayoutInflater.from(parent.context).inflate(R.layout.template_re_layout, parent, false)
        return ViewHolderClass(itemView)
    }

    override fun onBindViewHolder(holder: ViewHolderClass, position: Int) {
        val Elements = mList[position]
        holder.textView.setText(Elements.StudentNames)
    }

    override fun getItemCount(): Int {
        return mList.size
    }

    class ViewHolderClass(ItemView: View) : RecyclerView.ViewHolder(ItemView) {
        val textView: TextView = ItemView.findViewById(R.id.txtName)
    }
}
```

```
implementation ("androidx.lifecycle:lifecycle-viewmodel-ktx:2.6.1")
implementation ("androidx.activity:activity-ktx:1.7.2")
implementation ("androidx.cardview:cardview:1.0.0")
```

Week 7 – Correct in Recycler_Activity

Kotlin code

```
lateinit var lsStudents: RecyclerView
var strList: Array<String> = arrayOf("Nguyen Van Tien", "Bui Cong Minh",
"Nguyen Pha Le", "Phan Tuan Sy", "Dang Tieu Binh")
```

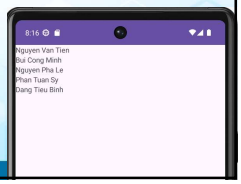
- **Binding data**

```
lsStudents= findViewById(R.id.lsStudents)
lsStudents.layoutManager = LinearLayoutManager(this)
lsStudents.setHasFixedSize(true )

val data = ArrayList<DataClass>()

for(i in 0 until  strList.size)
    data.add(DataClass(strList[i].toString()))

val adapter = RecyclerViewAdapter(data)
lsStudents.setAdapter(adapter)
```



template_re_layout layout: add ImageView

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"    >

    <ImageView
        android:layout_width="60dp"
        android:layout_height="60dp"
        android:id="@+id/imgPro"
        android:padding="8dp"
        android:scaleType="fitXY"
        android:src="@drawable/logo" />
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/txtName"    />

</LinearLayout>
```

Week 7 – DataClass ➔ ImageView by Int

```
data class DataClass(var imgProfire: Int, var StudentNames:String)
```

Week 7 – Adapter

```
override fun onBindViewHolder(holder: ViewHolderClass, position: Int) {
    val Elements = mList[position]
    holder.textView.setText(Elements.StudentNames)
    holder.image.setImageResource(Elements.imgProfire)
}

class ViewHolderClass(ItemView: View) : RecyclerView.ViewHolder(ItemView) {
    var image: ImageView = ItemView.findViewById(R.id.imgPro)
    val textView: TextView = ItemView.findViewById(R.id.txtName)
}
```

Week 7 –Recycler_Activity

- **Binding data**

```
data.add(DataClass(R.drawable.logo, strList[i].toString()))
```

